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Design and Analysis of Algorithm Digital Assignment - 1

① Activity selection

Set = {A1, A2, A3, A4, A5, A6, A7, A8, A9, A10}

Start Time = {3, 2, 4, 8, 7, 9, 1, 5, 6, 10}

End Time = {4, 3, 4, 10, 9, 10, 4, 7, 7, 11}

Sorting the activities in increasing order of their end times

Set: (A2) (A1) (A3) (A7) (A8) (A9) (A5) (A4) (A6) (A10)

Start time: 2 → 3 → 4 → 1 → 5 → 6 → 7 → 8 → 9 → 10

End time: 3 → 4 → 4 → 4 → 7 → 7 → 9 → 10 → 10 → 11

Selecting A2, then A1, then A3, but then A7 is interfering, then A8, but then A9 is interfering, then we take A5, then A4 is interfering, then A6 & A10

No. of activities executed are : 7
i.e. {A2, A1, A3, A8, A5, A6, A10}

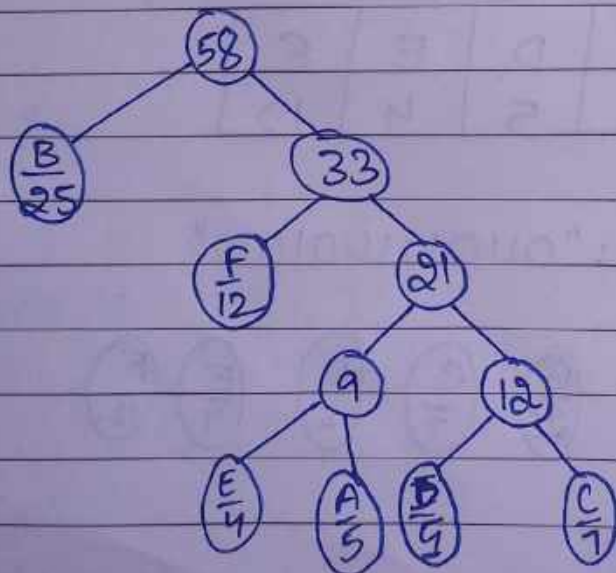
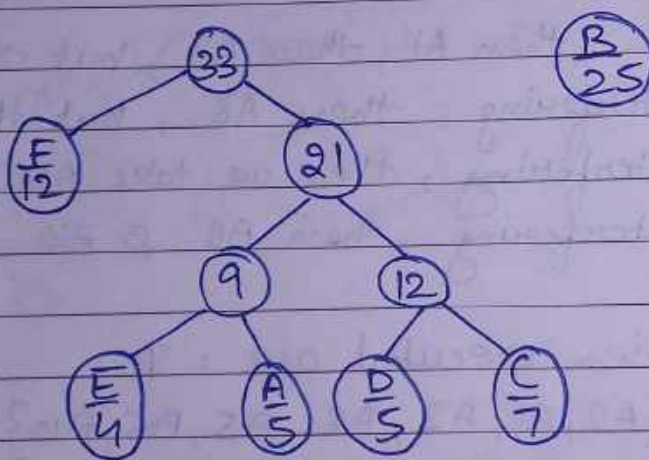
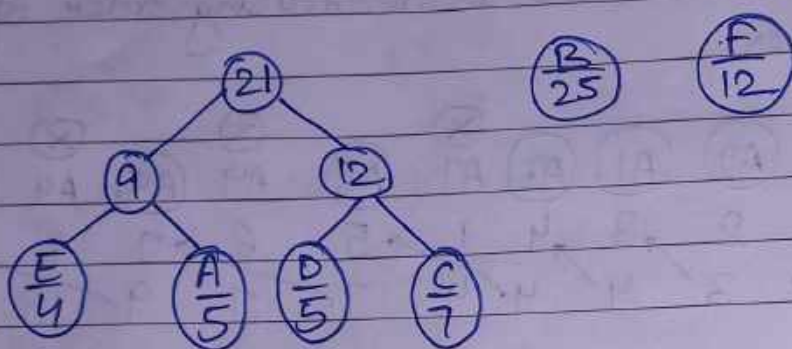
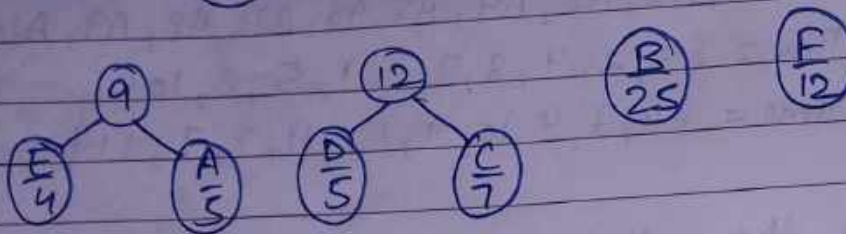
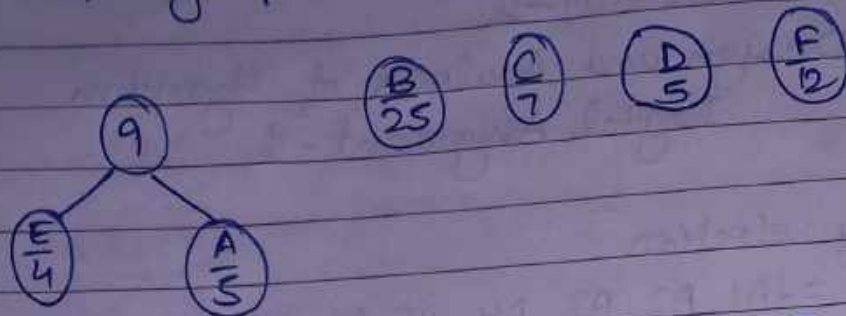
②

A	B	C	D	E	F
5	25	7	5	4	12

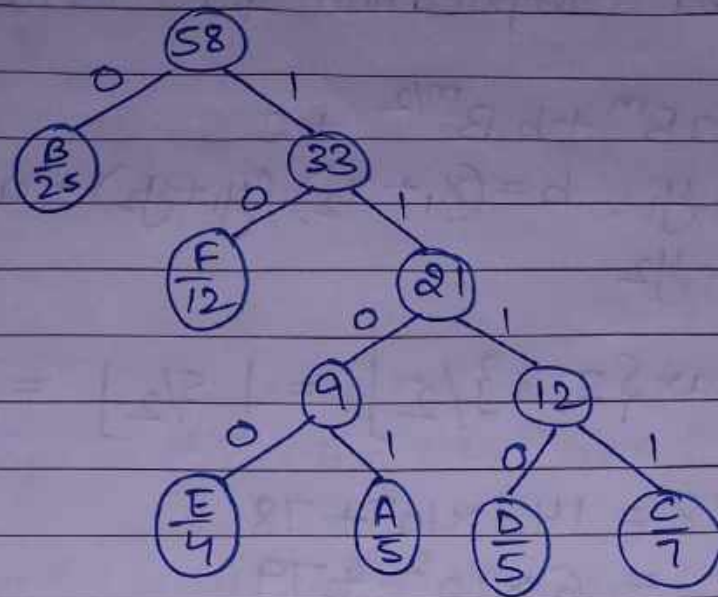
Message given: "0110111001010"

Nodes are: (A/5) (B/25) (C/7) (D/5) (E/4) (F/12)

Comparing frequencies!



Now labelling left child path as 0 and right as 1

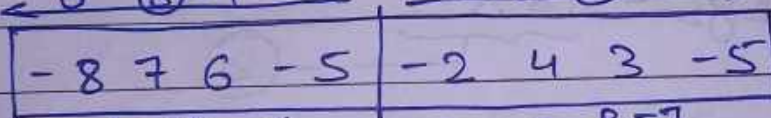


Message: 0110111001010
 B A E F F

Decoded string: **BAEFF** Ans

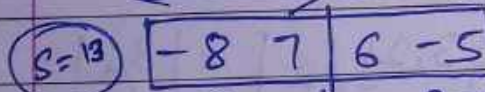
③ Max. subarray sum for $S = \{-8, 7, 6, -5, -2, 4, 3, -5\}$

← 0 ⑧ 1 -5 -2 2 ⑤ 0 →



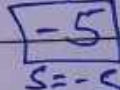
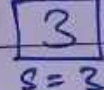
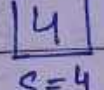
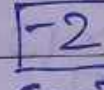
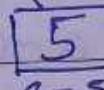
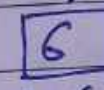
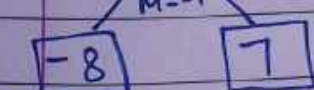
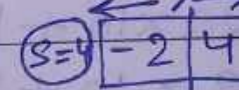
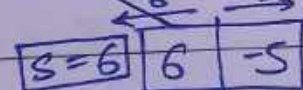
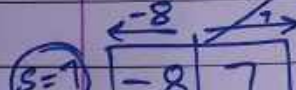
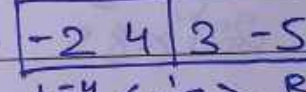
S=13

← -1 7 6 1 →



S=7

← 2 4 3 -2 →



Max. Subarray Sum = 13

Subarray is: $\{7, 6, -5, -2, 4, 3\}$

Ans

④ Karatsuba Multiplication for 14378×679

$$x \times y = aB^m + bB^{m/2} + c$$

$$a = x_1 \cdot y_1 \quad b = (x_1 + x_2) \cdot (y_1 + y_2) - a - c$$

$$c = x_2 \cdot y_2$$

$$m = \lfloor \max\{5, 3\}/2 \rfloor = \lfloor 5/2 \rfloor = 2$$

$$\therefore 14378 = 143 \times 10^2 + 78$$

$$679 = 6 \times 10^2 + 79$$

$$a = x_1 \cdot y_1 = 143 \times 6 = 858$$

$$b = (143 + 78)(6 + 79) - a - c$$

$$= \underbrace{221 \times 85 - a - c}_{\textcircled{1}}$$

$$c = x_2 \cdot y_2 = \underbrace{78 \times 79}_{\textcircled{2}}$$

Solving ①

$$221 \times 85$$

$$m = \lfloor \max\{3, 2\}/2 \rfloor = \lfloor 3/2 \rfloor = 1$$

$$221 = 22 \times 10 + 1$$

$$85 = 8 \times 10 + 5$$

$$a = x_1 \cdot y_1 = 22 \times 8 = 176$$

$$b = \underbrace{23 \times 13 - a - c}_{\textcircled{3}}$$

$$c = 1 \times 5 = 5$$

Solving ③

$$m = \lfloor \max\{2, 2\}/2 \rfloor = 1$$

$$23 = 2 \times 10 + 3$$

$$13 = 1 \times 10 + 3$$

$$a = 2 \times 1 = 2$$

$$b = 5 \times 4 - a - c$$

$$c = 9$$

$$\therefore b = 20 - 2 - 9$$

$$b = 9$$

$$\therefore 23 \times 13 = 2 \times 10^2 + 9 \times 10 + 9$$

$$= 200 + 90 + 9 = 299$$

$$\text{Putting in (3)} \therefore b = 299 - 176 - 5 = 118$$

$$\therefore 221 \times 85 = 176 \times 10^2 + 118 \times 10 + 5$$

$$= 18785$$

Solving (2)

$$78 \times 79 : m = \lfloor \max\{2, 2\} / 2 \rfloor = 1$$

$$78 = 7 \times 10 + 8$$

$$79 = 7 \times 10 + 9$$

$$a = 49 \quad b = \underbrace{15 \times 16}_{(4)} - a - c \quad c = 72$$

Solving (4)

$$15 = 1 \times 10 + 5$$

$$16 = 1 \times 10 + 6$$

$$a = 1 \quad b = 6 \times 7 - a - c \quad c = 30$$

$$= 42 - 1 - 30$$

$$b = 11$$

$$15 \times 16 = 100 + 21 \times 10 + 18$$

$$= 168 \quad 240$$

Putting in (4)

$$b = 240 - 49 - 72 = 119$$

$$\begin{aligned}
 78 \times 79 &= 49 \times 10^2 + 119 \times 10 + 72 \\
 &= 4900 + 1190 + 72 \\
 &= 6162
 \end{aligned}$$

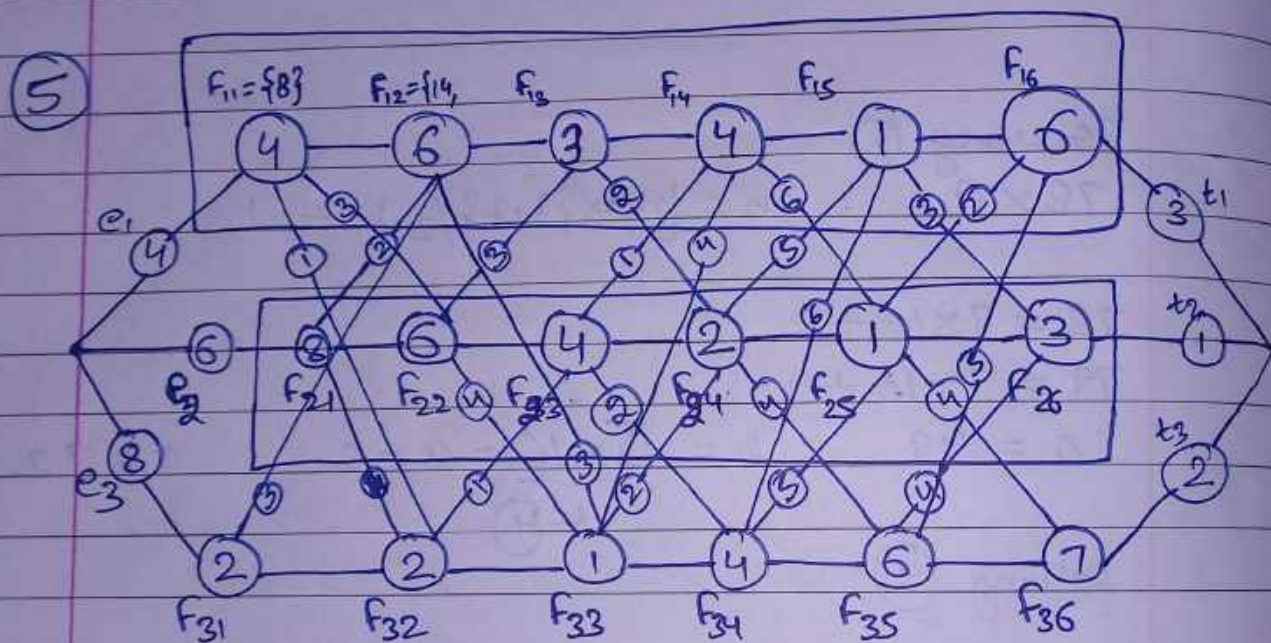
Put in ②

$$c = 6162$$

$$b = 18785 - 858 - 6162 = 11765$$

$$\therefore 14378 \times 679 = 858 \times 10^4 + 11765 \times 10^2 + 6162$$

$$= \boxed{9762662} \text{ Ans.}$$



$$F_{11} = 8$$

$$F_{12} = \min\{14, 22, 19\} = 14$$

$$F_{13} = \min\{17, 23, 15\} = 15$$

$$F_{14} = \min\{19, 21, 20\} = 19$$

$$F_{15} = \min\{20, 22, 23\} = 20$$

$$F_{16} = \min\{26, 25, 33\} = 25$$

$$F_{21} = 14$$

$$F_{22} = \min\{17, 20, 18\} = 17$$

$$F_{23} = \min\{20, 21, 16\} = 16$$

$$F_{24} = \min\{19, 18, 16\} = 16$$

$$F_{25} = \min\{26, 17, 22\} = 17$$

$$f_{26} = \min \{26, 20, 24\} = 20$$

$$f_{31} = 10$$

$$f_{32} = \min \{10, 20, 12\} = 11$$

$$f_{33} = \min \{18, 22, 12\} = 12$$

$$f_{34} = \min \{22, 22, 16\} = 16$$

$$f_{35} = \min \{30, 26, 22\} = 22$$

$$f_{36} = \min \{30, 28, 24\} = 28$$

So, the min cost can be represented as

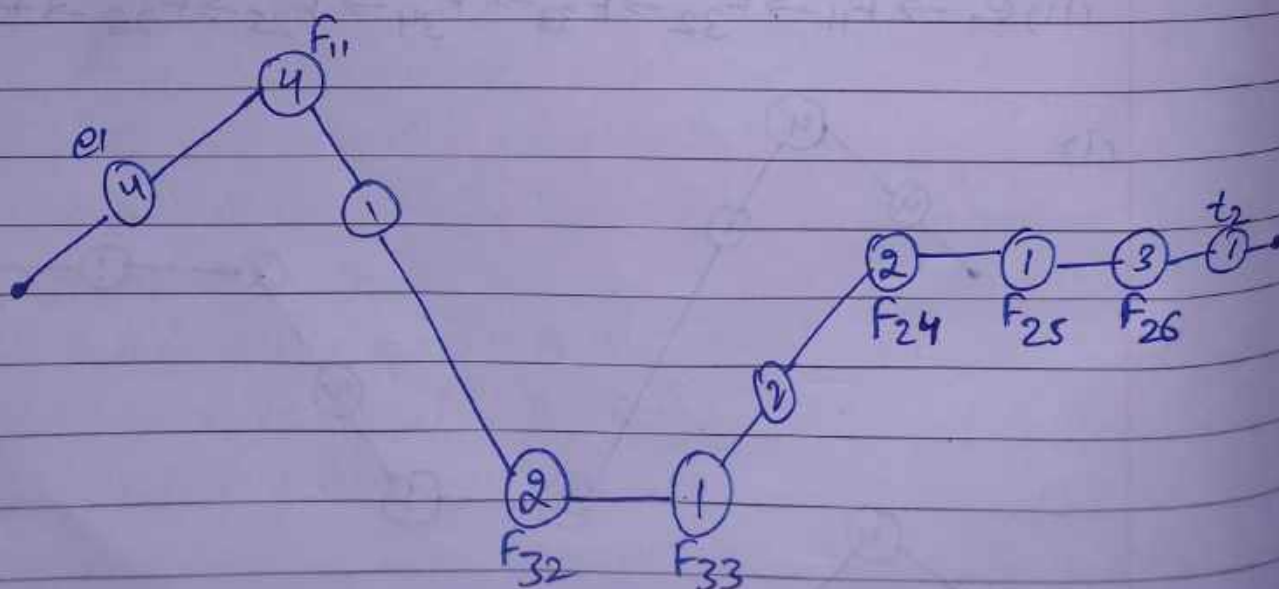
Station : 1 2 3 4 5 6

line:

	1	2	3	4	5	6
1	8	14	15	19	20	25
2	14	17	16	16	17	20
3	10	11	12	16	22	28

$\therefore$ Sequence : $e_1 \rightarrow f_{11} \rightarrow f_{32} \rightarrow f_{33} \rightarrow f_{24} \rightarrow f_{25} \rightarrow f_{26} \rightarrow t_2$

Total cost = 21



⑥ Matrix A1 A2 A3 A4 A5 A6
 Dimensions 10x4 4x2 2x3 3x5 5x6 6x5

P 10 | 4 | 2 | 3 | 5 | 6 | 5

	1	2	3	4	5	6		1	2	3	4	5	6
1	0	80	140	210	240	230	1	1	2	3	4	5	6
2		0	24	70	138	190	2		2	2	2	2	2
3			0	30	90	150	3			3	3	4	5
4				0	90	180	4				4	4	5
5					0	150	5					5	5
6						0	6						6

Step 1 Matrix:

$$M(1,1) = M(2,2) = M(3,3) = M(4,4) = M(5,5) = M(6,6) = 0$$

$$M(i,j) = \begin{cases} \min \{ M(i,k) + M(k+1,j) + P_{i-1} \cdot P_k \cdot P_j \} & \text{if } i < j \\ 0 & \text{if } i = j \end{cases}$$

Step 2 Matrix:

$$M(1,2): A_1 \times A_2 \therefore k=1: M(1,1) + M(2,2) + P_0 P_1 P_2 \\ = 0 + 0 + (10)(4)(2) \\ = 80$$

$$M(2,3): A_2 \times A_3 \therefore k=2: M(2,2) + M(3,3) + P_1 P_2 P_3 \\ = 0 + 0 + 4(2)(3) \\ = 24$$

$$M(3,4): A_3 \times A_4 \therefore k=3: 0 + 0 + 30 = 30$$

$$M(4,5): A_4 \times A_5 \therefore k=4: 0 + 0 + 90 = 90$$

$$M(5,6): A_5 \times A_6 \therefore k=5: 0 + 0 + 150 = 150$$

Step 3 Matrix:

$$M(1,3) \therefore k=1, 2$$

$$\begin{aligned}
 k=1 \quad M(1,3) &= M(1,1) + M(2,3) + P_0 P_1 P_3 \\
 &= 0 + 24 + 10(4)(3) \\
 &= 144
 \end{aligned}$$

$$\begin{aligned}
 k=2 \quad M(1,3) &= M(1,2) + M(3,3) + P_0 P_2 P_3 \\
 &= 80 + 0 + 60 \\
 &= 140
 \end{aligned}$$

$$M(1,3) = \min\{144, 140\} = \underline{140}$$

$$M(2,4) = k = 2, 3$$

$$\begin{aligned}
 k=2 \quad M(2,4) &= M(2,2) + M(3,4) + P_1 P_2 P_4 \\
 &= 0 + 30 + 40 \\
 &= 70
 \end{aligned}$$

$$\begin{aligned}
 k=3 \quad M(2,4) &= M(2,3) + M(4,4) + P_1 P_3 P_4 \\
 &= 24 + 0 + 60 \\
 &= 84
 \end{aligned}$$

$$M(2,4) = \min\{70, 84\} = \underline{70}$$

$$M(3,5) \quad \therefore k = 3, 4$$

$$\begin{aligned}
 k=3 \quad M(3,5) &= M(3,3) + M(4,5) + P_2 P_3 P_5 \\
 &= 0 + 90 + 36 \\
 &= 126
 \end{aligned}$$

$$\begin{aligned}
 k=4 \quad M(3,5) &= M(3,4) + M(5,5) + P_2 P_4 P_5 \\
 &= 30 + 0 + 60 \\
 &= 90
 \end{aligned}$$

$$M(3,5) = \min\{90, 126\} = \underline{90}$$

$$M(4,6) \therefore k=4,5$$

$$\begin{aligned} k=4 \quad M(4,6) &= M(4,4) + M(5,6) + P_3 P_4 P_6 \\ &= 0 + 150 + 70 \\ &= 225 \end{aligned}$$

$$\begin{aligned} k=5 \quad M(4,6) &= M(4,5) + M(6,6) + P_3 P_5 P_6 \\ &= 90 + 0 + 90 \\ &= 180 \end{aligned}$$

$$M(4,6) = \min\{225, 180\} = \underline{180}$$

Step 4 4 Matrix

$$M(1,4) \therefore k=1,2,3$$

$$\begin{aligned} k=1 \quad M(1,4) &= M(1,1) + M(2,4) + P_0 P_1 P_4 \\ &= 0 + 70 + 200 = 270 \end{aligned}$$

$$\begin{aligned} k=2 \quad M(1,4) &= M(1,2) + M(3,4) + P_0 P_2 P_4 \\ &= 80 + 30 + 100 = 210 \end{aligned}$$

$$\begin{aligned} k=3 \quad M(1,4) &= M(1,3) + M(4,4) + P_0 P_3 P_4 \\ &= 140 + 0 + 150 = 290 \end{aligned}$$

$$M(1,4) = \min\{270, 210, 290\} = \underline{210}$$

$$M(2,5) \therefore k=2,3,4$$

$$\begin{aligned} k=2 \quad M(2,5) &= M(2,2) + M(3,5) + P_1 P_2 P_5 \\ &= 0 + 90 + 48 = 138 \end{aligned}$$

$$\begin{aligned} k=3 \quad M(2,5) &= M(2,3) + M(4,5) + P_1 P_3 P_5 \\ &= 24 + 90 + 72 \\ &= 186 \end{aligned}$$

$$\begin{aligned} k=4 \quad M(2,5) &= M(2,4) + M(5,5) + P_1 P_4 P_5 \\ &= 70 + 0 + 120 \\ &= 190 \end{aligned}$$

$$M(2,5) = \min\{138, 186, 190\} = \underline{138}$$

$$M(3,6) \therefore k=3,4,5$$

$$k=3 \quad M(3,6) = M(3,3) + M(4,6) + P_2 P_3 P_6 \\ = 0 + 180 + 30 = 210$$

$$k=4 \quad M(3,6) = M(3,4) + M(5,6) + P_2 P_4 P_6 \\ = 30 + 150 + 50 = 230$$

$$k=5 \quad M(3,6) = M(3,5) + M(6,6) + P_2 P_5 P_6 \\ = 90 + 0 + 60 = 150$$

$$M(3,6) = \min \{210, 230, 150\} = \underline{150}$$

Steps S Matrix

$$M(1,5) \therefore k=1,2,3,4$$

$$k=1 \quad M(1,5) = 0 + 138 + 240 \\ = 378$$

$$k=2 \quad M(1,5) = M(1,2) + M(3,5) + P_0 P_2 P_5 \\ = 80 + 90 + 120 \\ = 290$$

$$k=3 \quad M(1,5) = M(1,3) + M(4,5) + P_0 P_3 P_5 \\ = 140 + 90 + 180 \\ = 410$$

$$k=4 \quad M(1,5) = 210 + 0 + 300 \\ = 510$$

$$M(1,5) = \min \{378, 290, 410, 510\} = \underline{290}$$

$$M(2,6) \therefore k=2,3,4,5$$

$$k=2 \quad M(2,6) = M(2,2) + M(3,6) + P_1 P_2 P_6 \\ = 0 + 150 + 40 \\ = 190$$

$$k=3 \quad M(2,6) = 24 + 150 + 60 \\ = 234$$

$$k=4 \quad M(2,6) = 20 + 150 + 100 \\ = 370$$

$$k=5 \quad M(2,6) = M(2,5) + M(6,6) + P_1 P_5 P_6 \\ = 138 + 0 + 120 \\ = 258$$

$$M(2,6) = \min\{190, 234, 370, 258\} \\ = \underline{190}$$

Step 6 $M(1,6) \therefore k=1,2,3,4,5,$

$$k=1 \quad M(1,6) = M(1,1) + M(2,6) + P_0 P_1 P_6 \\ = 0 + 190 + 200 \\ = 390$$

$$k=2 \quad M(1,6) = M(1,2) + M(3,6) + P_0 P_2 P_6 \\ = 80 + 150 + 100 \\ = 330$$

$$k=3 \quad M(1,6) = M(1,3) + M(4,6) + P_0 P_3 P_6 \\ = 140 + 180 + 150 \\ = 470$$

$$k=4 \quad M(1,6) = 210 + 150 + 250 \\ = 610$$

$$k=5 \quad M(1,6) = M(1,5) + M(6,6) + P_0 P_5 P_6 \\ = 290 + 0 + 300 \\ = 590$$

$$M(1,6) = \min\{390, 330, 470, 610, 590\}$$

$$= \boxed{330}$$

$$\boxed{(A_1 \times A_2) \times (((A_3 \times A_4) \times A_5) \times A_6)}$$

Ans.

⑦ Longest Common Subsequence
 "ATGGATBBT" & "GTABTABAT"

		A	T	G	G	A	T	B	B	T	0
0	0	0	0	0	0	0	0	0	0	0	0
G	0	0	0	1	1	1	1	1	1	1	1
T	0	0	1	1	1	1	2	2	2	2	2
A	0	1	1	1	1	2	2	2	2	2	2
B	0	1	1	1	1	2	2	3	3	3	3
T	0	1	2	2	2	2	3	3	3	4	4
A	0	1	2	2	2	3	3	3	3	4	4
B	0	1	2	2	2	3	3	4	4	4	4
A	0	1	2	2	2	3	3	4	4	4	4
T	0	1	2	2	2	3	4	4	4	5	5
0	0	1	2	2	2	3	4	4	4	5	5

↓	↓					↓			↓	↓
①	A	T				A			B	T
②		T				A		B	B	T
③		T				A	T	B		T
④			G			A	T	B		T
⑤			G				T	B	B	T

So, total five longest common subsequences are possible i.e.

ATABT	}	Ans
TABBT		
TATBT		
GATBT		
GTBBT		

⑧ 0/1 Knapsack

Item	Profit	Weight
1	6	2
2	5	4
3	2	5
4	4	1
5	3	3

$$W = 9$$

	0	1	2	3	4	5	6	7	8	9
Item	0	0	0	0	0	0	0	0	0	0
1	0	0	6	6	6	6	6	6	6	6
2	0	0	6	6	6	6	11	11	11	11
3	0	0	6	6	6	6	11	11	11	11
4	0	4	6	10	10	10	11	15	15	15
5	0	4	6	10	10	10	13	15	15	15

$$\{1, 1, 0, 1, 0\} \quad \text{Ans.}$$

i.e. we will take 1st, 2nd & 4th item.

⑨ Travelling Salesman Problem

	A	B	C	D	E
A	0	2	12	4	8
B	2	0	5	7	3
C	12	5	0	10	1
D	4	7	10	0	4
E	8	3	1	4	0

No intermediate:

$$C_{B,\phi,A} = 2$$

$$C_{C,\phi,A} = 12$$

$$C_{D,\phi,A} = 4$$

$$C_{E,\phi,A} = 8$$

1 intermediate:

$$C_{B,(C),A} = d_{B,C} + C_{C,\phi,A} = 5 + 12 = 17$$

$$C_{B,(D),A} = d_{B,D} + C_{D,\phi,A} = 7 + 4 = 11$$

$$C_{B,(E),A} = d_{B,E} + C_{E,\phi,A} = 3 + 8 = 11$$

$$C_{C,(B),A} = d_{C,B} + C_{B,\phi,A} = 5 + 2 = 7$$

$$C_{C,(D),A} = d_{C,D} + C_{D,\phi,A} = 10 + 4 = 14$$

$$C_{C,(E),A} = d_{C,E} + C_{E,\phi,A} = 1 + 8 = 9$$

$$C_{D,(B),A} = d_{D,B} + C_{B,\phi,A} = 7 + 2 = 9$$

$$C_{D,(C),A} = d_{D,C} + C_{C,\phi,A} = 10 + 12 = 22$$

$$C_{D,(E),A} = d_{D,E} + C_{E,\phi,A} = 4 + 8 = 12$$

$$C_{E,(B),A} = d_{E,B} + C_{B,\phi,A} = 3 + 2 = 5$$

$$C_{E,(C),A} = d_{E,C} + C_{C,\phi,A} = 1 + 12 = 13$$

$$C_{E,(D),A} = d_{E,D} + C_{D,\phi,A} = 4 + 4 = 8$$

2 intermediates:

$$\begin{aligned} C_{B,(C,D),A} &= \min\{d_{BC} + C_{CDA}, d_{BD} + C_{DCA}\} \\ &= \min\{5 + 14, 7 + 22\} \\ &= \min\{19, 29\} \\ &= 19 \end{aligned}$$

$$\begin{aligned} C_{B,(C,E),A} &= \min\{d_{BC} + C_{CEA}, d_{BE} + C_{ECA}\} \\ &= \min\{14, 16\} \\ &= 14 \end{aligned}$$

$$\begin{aligned} C_{B,(D,E),A} &= \min\{d_{BD} + C_{DEA}, d_{BE} + C_{EDA}\} \\ &= \min\{14, 11\} \\ &= 11 \end{aligned}$$

$$C_{C,(B,D),A} = \min \{ d_{CB} + C_{BDA}, d_{CD} + C_{CBA} \}$$

$$= \min (16, 19)$$

$$= 16$$

$$C_{C,(B,E),A} = \min \{ d_{CB} + C_{BEA}, d_{CE} + C_{CBA} \}$$

$$= \min (16, 6)$$

$$= 6$$

$$C_{C,(E,D),A} = \min \{ d_{CE} + C_{EDA}, d_{CD} + C_{CEA} \}$$

$$= \min (9, 22)$$

$$= 9$$

$$C_{D,(B,C),A} = \min \{ d_{DB} + C_{BCA}, d_{DC} + C_{CBA} \}$$

$$= \min \{ 24, 17 \}$$

$$= 17$$

$$C_{D,(B,E),A} = \min \{ d_{DB} + C_{BEA}, d_{DE} + C_{CBA} \}$$

$$= \min \{ 18, 9 \}$$

$$= 9$$

$$C_{D,(C,E),A} = \min \{ d_{DC} + C_{CEA}, d_{DE} + C_{ECA} \}$$

$$= \min \{ 19, 17 \}$$

$$= 17$$

$$C_{E,(B,C),A} = \min \{ d_{EB} + C_{BCA}, d_{EC} + C_{CBA} \}$$

$$= \min \{ 20, 8 \}$$

$$= 8$$

$$C_{E,(B,D),A} = \min \{ d_{EB} + C_{BDA}, d_{ED} + C_{DBD} \}$$

$$= \min \{ 14, 13 \}$$

$$= 13$$

$$C_{E,(C,D),A} = \min \{ d_{EC} + C_{CDA}, d_{ED} + C_{DCA} \}$$

$$= \min \{ 15, 26 \}$$

$$= 15$$

3 intermediates:

$$C_{B,(C,D,E),A} = \min \{ d_{BC} + C_{(D,E),A}, d_{BD} + C_{B(C,E),A}, d_{BE} + C_{B(CD),A} \}$$

$$= \min \{ 14, 24, 18 \}$$

$$= 14$$

$$C_C(BDE), A = \min \{ 16, 19, 14 \}$$

$$= 14$$

$$C_D(BCE), A = \min \{ d_{DB} + C_C(BE), A, d_{DC} + C_C(BE), A, d_{DE} + C_C(BC), A \}$$

$$= \min \{ 21, 16, 12 \}$$

$$= 12$$

$$C_E(B, CD), A = \min \{ d_{EB} + C_D(CD), A, d_{EC} + C_D(CD), A, d_{ED} + C_D(BC), A \}$$

$$= \{ 22, 22, 21 \}$$

$$= \textcircled{21}$$

4 intermediates:

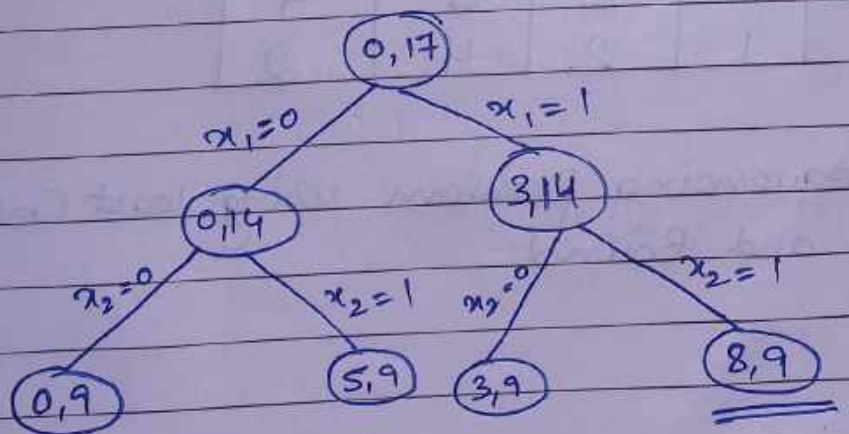
$$C_A(BDE), A = \min \begin{cases} d_{AB} + C_B(DE), A = 16 \\ d_{AC} + C_C(BDE), A = 26 \\ d_{AD} + C_D(BE), A = 16 \\ d_{AE} + C_E(BCD), A = 29 \end{cases}$$

$\therefore$ Min distance = 16

Path 1: ABCEDA } Ans

Path 2: ADECB

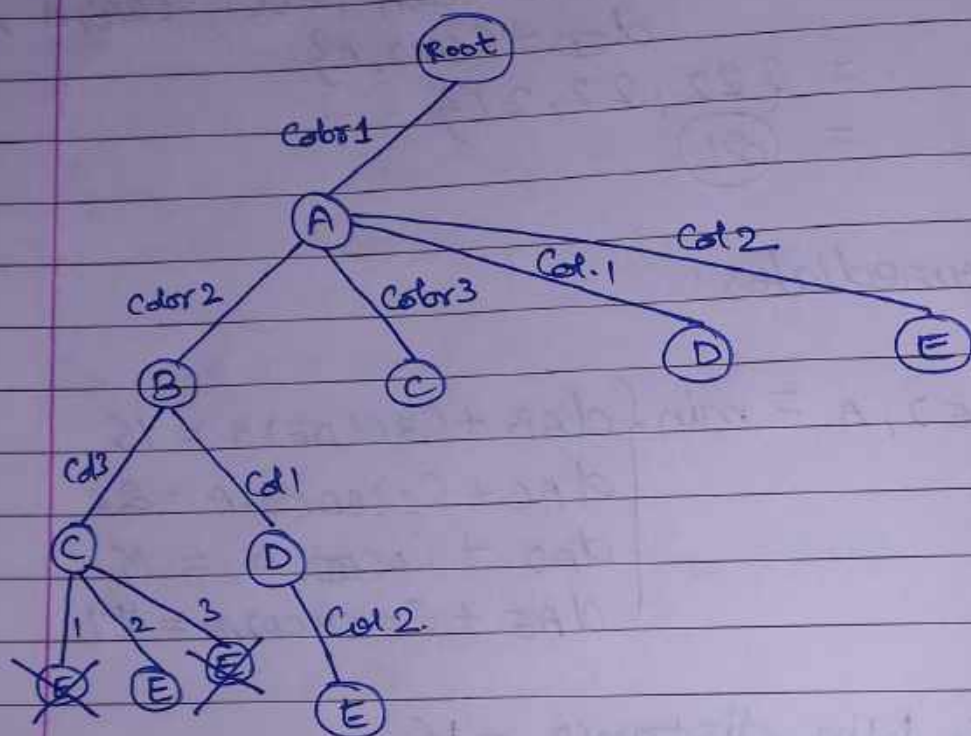
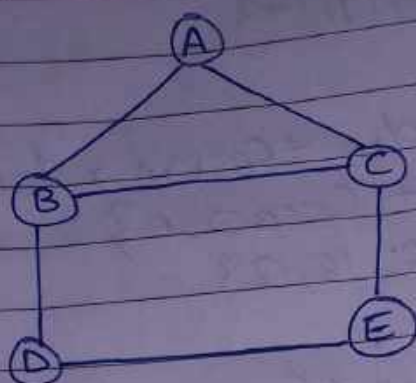
⑩ $n=5$ $W=\{3, 5, 2, 3, 4\}$ $\text{Sum}=8$



Subset: $\{1, 1, 0, 0, 0\}$
Sum=8

Ans

(11)

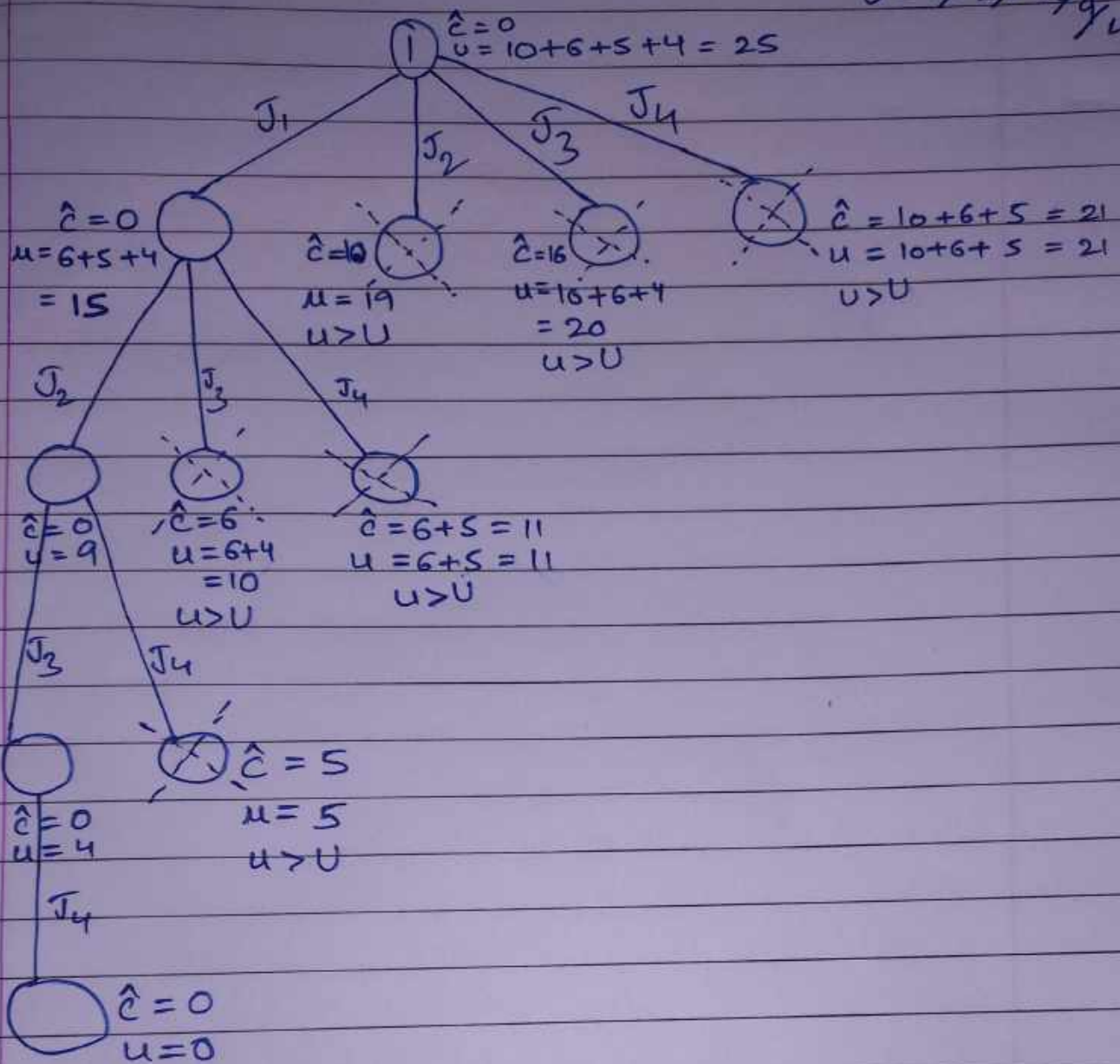


$\therefore$ No. of colors required = 3

(12)

Job	1	2	3	4
Profit	10	6	5	3
Deadline	2	2	3	4
Time	1	2	4	2

Job Sequencing problem using least Cost Branch and Bound.



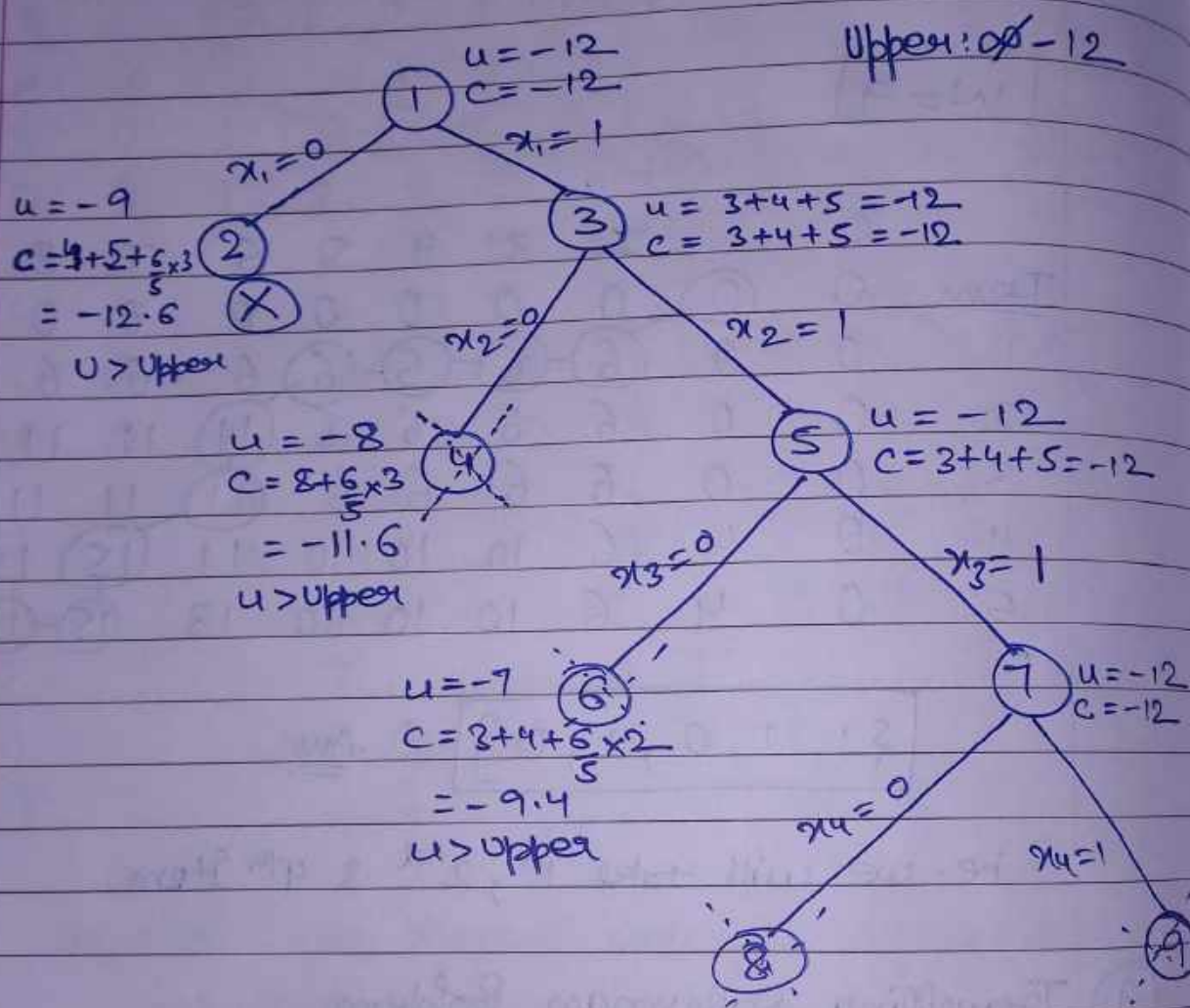
$\therefore$ The schedule for jobs is: Job1, Job2, Job3, Job4

Total profit = $10+6+5+3$

$= 24$

12) Branch and Bound o/l Knapsack

W=8	Profit	Weight	Item
	3	3	1
	4	3	2
	5	2	3
	6	5	4



Since $w=8$ is reached at 7th node
So we can't take 4th item.

$$\therefore \text{Ans} = \boxed{\{1, 1, 1, 0\}}$$